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## CHAPTER 1 • NUMBER • TOPIC 12

# Ratio Toolkit

A ratio is parts. Find the value of one part, then build every answer from it — scale, share, link and solve.



$$\frac{3}{10} = 0.3 =$$

**30%****5**

VARIANTS

**5**

PAST-PAPER PARTS

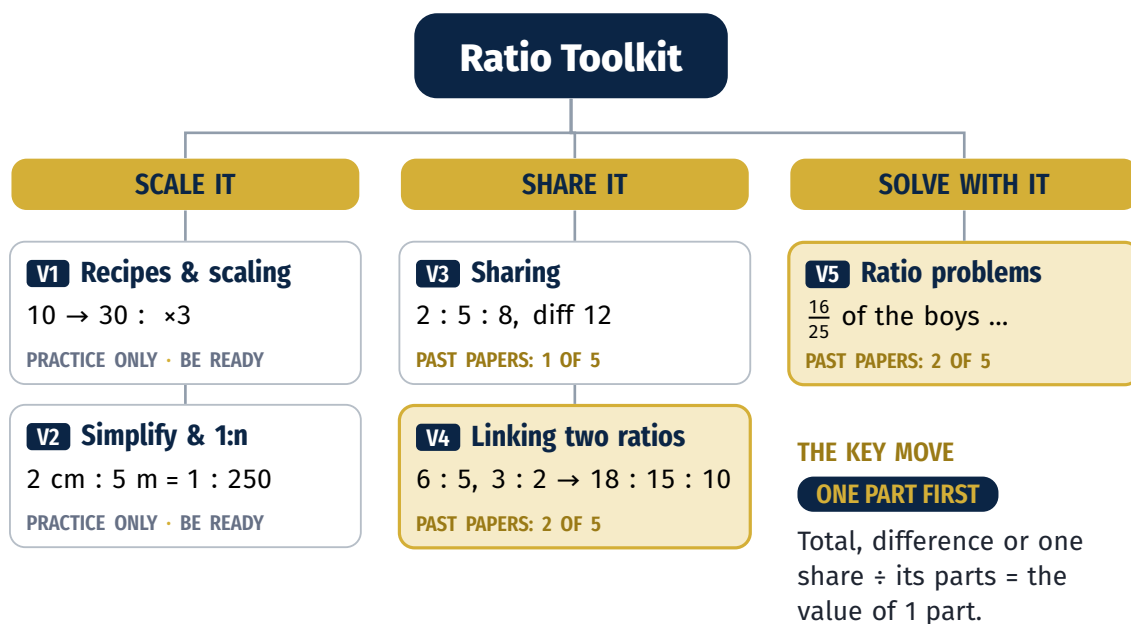
**55**

PRACTICE QUESTIONS

**YOU WILL MASTER****V1** Recipes & scaling**V2** Simplify & 1:n**V3** Sharing**V4** Linking two ratios**V5** Ratio problems

## TOPIC MAP • EVERY VARIANT ON ONE PAGE

## Ratio at a glance


  
Your notes


### ★ How often does each variant come up?

5 past-paper questions (Jan 2020  $\rightarrow$  Nov 2023); 55 practice parts — grouped by type. More ratio questions are in Topic 13.

|                              | PAST PAPERS (OUT OF 5) |   | PRACTICE PARTS                   |    |
|------------------------------|------------------------|---|----------------------------------|----|
| <b>V1</b> Recipes, scaling   | ● ● ● ● ●              | 0 | <div style="width: 100%;"></div> | 13 |
| <b>V2</b> Simplify and 1:n   | ● ● ● ● ●              | 0 | <div style="width: 50%;"></div>  | 8  |
| <b>V3</b> Sharing            | ● ● ● ● ●              | 1 | <div style="width: 100%;"></div> | 15 |
| <b>V4</b> Linking two ratios | ● ● ● ● ●              | 2 | <div style="width: 20%;"></div>  | 2  |
| <b>V5</b> Ratio problems     | ● ● ● ● ●              | 2 | <div style="width: 100%;"></div> | 19 |

**WHAT THIS TELLS YOU** The exam skips the easy scaling and goes straight to linking two ratios and multi-step stories (3–6 marks, early in Paper 1 or 2). Every one is cracked by finding one part first.

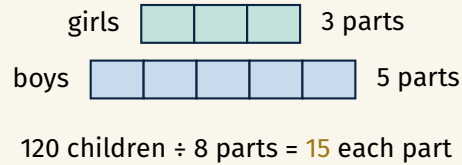
## BEFORE YOU START

## The big idea

**A ratio is a number of equal parts.**

Girls : boys = 3 : 5 means for every 3 girls there are 5 boys — **8 parts** in total.

Once you know what **one part** is worth, everything else is multiplying.


 **The four numbers to find**
**Parts**

$$3 : 5$$

Each number is a count of equal parts.

**Total**

$$3 + 5 = 8$$

Add the parts when a total is given.

**Difference**

$$5 - 3 = 2$$

Subtract the parts when “more than” is given.

**One part**

$$120 \div 8 = 15$$

Divide what you know by its parts.

 **Which number do you divide by?**
**“share £120”**

$$120 \div (3 + 5)$$

A total: divide by the total parts.

**“12 more than”**

$$12 \div (5 - 2)$$

A difference: divide by the difference.

**“120 g of sugar”**

$$120 \div 2$$

One share: divide by its own parts.

**CALCULATOR FIRST, THEN CHECK**

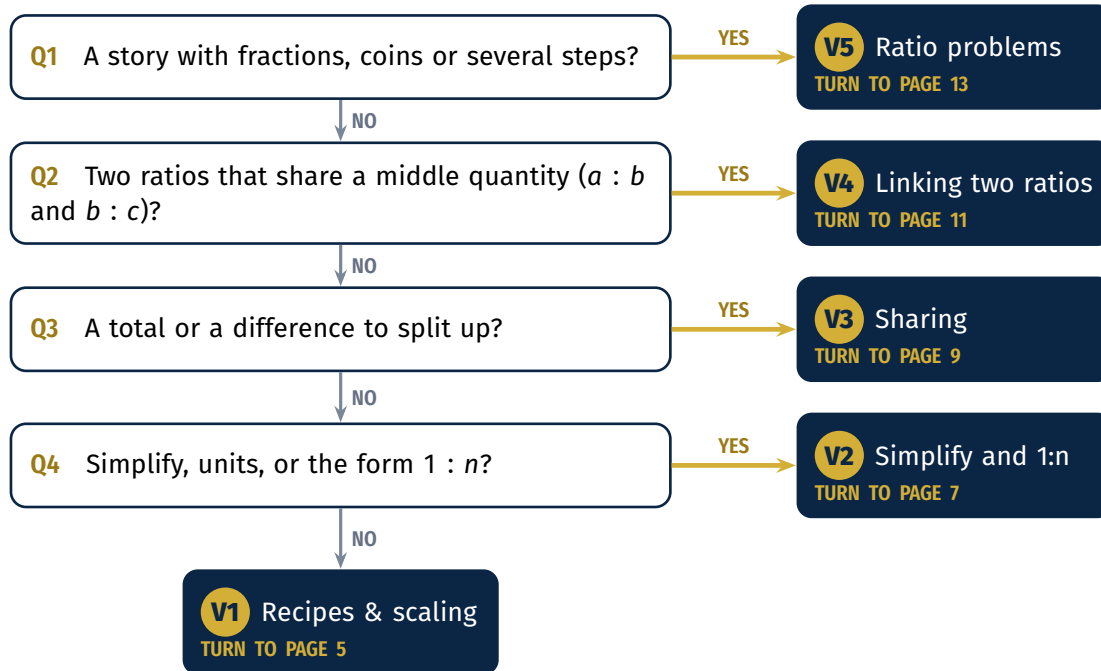
Check the answer: the shares must add to the total and still be in the given ratio (45 : 75 = 3 : 5 ✓).

COUNT / RECOGNISE, THEN CHOOSE

# Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



## COLOUR CODE IN THE WORKING

**Blue** = letter terms  $4x$ ,  $-3y$

**Orange** = plain numbers  $17$ ,  $-13$

**STEP 1** **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

## ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1

VARIANT 1

# Recipes & Scaling

Find the multiplier (new  $\div$  old) and use it on every amount.

SEEN IN PAST PAPERS

● ● ● ● ● 0/5

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 13


  
Your notes

## ? What is it?

### In plain words

A recipe makes a fixed number. To make more or fewer, find the **multiplier**: new  $\div$  old.

Multiply **every ingredient** by it.  
“Greatest number you can make”: find each ingredient’s multiplier — the **smallest** one wins.

### IT LOOKS LIKE

10 pancakes  $\rightarrow$  30

enough flour?

greatest number

10 pancakes  
120 g flour

$\times 3$  (30  $\div$  10)

30 pancakes  
360 g flour

## How do I do it?

### 1 Multiplier

new  $\div$  old (people, cakes, biscuits).

### 2 Scale

Every ingredient  $\times$  the multiplier.

### 3 Limit

Greatest number: have  $\div$  need for each — take the smallest.

### 4 Answer

Units; “enough?” — compare and say Yes/No.

**NEW  $\div$  OLD = MULTIPLIER** Same multiplier for every ingredient.

**V1 Recipes & Scaling**

VARIANT 1 • SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO***"Ingredients for 4 people ... for 6 people"*→ **Multiplier**  $6 \div 4 = 1.5$ *"Does she have enough?"*→ **Work out what she needs, compare, then Yes/No***"The greatest number he can make"*→ **Have ÷ need for each; smallest wins****WORKED EXAMPLE****PRACTICE**

Practice • Easy • Q1 • 4 marks

10 pancakes need 300 ml milk, 120 g flour, 2 eggs. (a) How much flour for 30 pancakes? (b) Tara uses 750 ml of milk. How many pancakes?

**STEP 1**

$30 \div 10 = 3$

◀ (a) multiplier

**STEP 2**

$120 \times 3 = 360 \text{ g}$

◀ scale the flour

**STEP 3**

$750 \div 300 = 2.5$

◀ (b) multiplier from the milk

**STEP 4**

$10 \times 2.5 = 25 \text{ pancakes}$

◀ scale the pancakes

**ANSWER** (a) 360 g (b) 25 pancakes

✓ 25 pancakes need 750 ml ✓

**WATCH OUT**

$\times 120 + 20 = 140 \text{ g}$

$\checkmark 120 \times 3 = 360 \text{ g}$

Recipes scale by multiplying, never by adding.

**EXAM TIP**

"Enough?" is a show question: write what is needed, what she has, then Yes or No.

**Check yourself**

calculator first • answers on page 15

**1.1**

Practice • Easy

16 gingerbread men:  
180 g flour, 40 g ginger,  
110 g butter, 30 g sugar.  
Amounts for 24?

Answer: .....

**1.2**

Practice • Easy

225 g flour makes 9  
cakes. Is 475 g enough  
for 20 cakes?

Answer: .....

**1.3**

Practice • Medium

8 biscuits: 120 g butter,  
60 g sugar, 180 g flour.  
He has 330 g, 200 g, 450  
g. Greatest number?

Answer: .....



## VARIANT 2

## VARIANT 2

# Simplify & the Form

## 1:n

Same units, divide by the HCF — or divide by the first number for 1:n.

## SEEN IN PAST PAPERS

0 / 5

**NOT ASKED YET • BE READY**

## PRACTICE QUESTIONS: 8



Your notes

## What is it?

### In plain words

**Simplify:** divide every part by the same number (the HCF), like a fraction.

**Units first:** 2 cm : 5 m must become 2 : 500. Form **1 : n**: divide both parts by the first number.

## IT LOOKS LIKE

$$24 : 36 : 60$$
$$50 \text{ g} : 1 \text{ kg} = 1 : n$$
$$2 \text{ cm} : 5 \text{ m}$$

### SAME UNITS

2 : 500

÷2 BOTH

1 : 250

## How do I do it?

## 1 Units

Change to the smaller unit: m  $\rightarrow$  cm, kg  $\rightarrow$  g.

## 2 Divide

Every part by the HCF (simplest form).

**3 1:n**

Divide every part by the first number.

## 4 Check

Whole numbers for “simplest”;  $n$  may be a decimal.

**SAME UNITS FIRST** A ratio has no units once simplified.

**V2 Simplify & the Form 1:n**

VARIANT 2 • SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO***"Write in its simplest form"*→ **Divide by the HCF until nothing divides all parts***"In the form 1 : n"*→ **Divide everything by the first number***Mixed units: cm and m, g and kg*→ **Convert first****WORKED EXAMPLE** **PRACTICE**

Practice • Easy • Q10(a) • 2 marks

The ratio 2 centimetres to 5 metres can be written in the form  $1 : n$ . Find the value of  $n$ .

**STEP 1**     5 m = 500 cm     ◀ same units

**STEP 2**     2 : 500     ◀ the ratio

**STEP 3**     1 : 250     ◀  $\div 2$  both

**ANSWER**  $n = 250$  $\checkmark 2 \times 250 = 500 \checkmark$ **WATCH OUT** $\times 2 \text{ cm} : 5 \text{ m} = 2 : 5$  $\checkmark 2 : 500 = 1 : 250$ 

Different units make the ratio wrong by a factor of 100.

**EXAM TIP**

$1 : n$  is ideal for maps and scales — Topic 13 and scale drawings use it.

**Check yourself**

calculator first • answers on page 15

**2.1** Practice • Easy

The ratio 50 grams to 1 kilogram can be written as  $1 : n$ . Find  $n$ .

Answer: .....

**2.2** Practice • Easy

A is half of B. Work out the ratio  $A : B$ .

Answer: .....

**2.3** Dr Eslam

Simplify  $24 : 36 : 60$ .

Answer: .....

**V3**

VARIANT 3

# Sharing: the Value of One Part

Divide the total (or the difference) by its number of parts, then multiply.

SEEN IN PAST PAPERS

● ● ● ● ● 1 / 5

PRACTICE QUESTIONS: 15

 Your notes

Came up in: May 2023 P1H Q1

## ? What is it?

### In plain words

Given a **total**:  $\text{total} \div (\text{sum of the parts}) = \text{one part}$ .

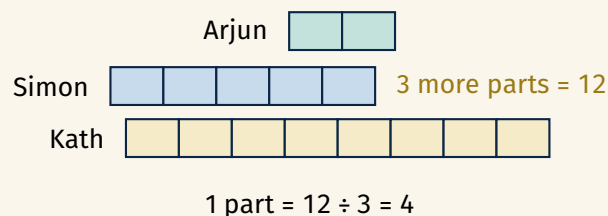
Given a **difference** ("12 more than"):  $\text{difference} \div (\text{difference of the parts}) = \text{one part}$ .

Then each share =  $\text{its parts} \times \text{one part}$ .

### IT LOOKS LIKE

share £630 in 5 : 13

12 more goals



## How do I do it?

### 1 Parts

Write the ratio and count the parts.

### 2 Match

Total ↔ sum of parts; difference ↔ difference of parts.

### 3 One part

Divide.

### 4 Shares

Multiply each number in the ratio by one part.

**ONE PART FIRST** Then every share is a multiplication.

**V3 Sharing: the Value of One Part**

VARIANT 3 • SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO**

"Share £630 in the ratio 5 : 13" →  $630 \div 18 = 35$  **per part**

"Simon scored 12 more goals than Arjun" →  $12 \div (5 - 2)$  **per part**

"How much does Harry receive?" → **Only his share: his parts × one part**

**WORKED EXAMPLE** **PAST PAPER**

May 2023 • Paper 1H • Q1 • 3 marks

Goals scored by Arjun, Simon and Kath were in the ratios 2 : 5 : 8. Simon scored 12 more goals than Arjun. Work out the number of goals scored by Kath.

**STEP 1**    2 : 5 : 8    ◀ parts

**STEP 2**     $5 - 2 = 3$  parts = 12    ◀ difference ↔ difference

**STEP 3**    1 part =  $12 \div 3 = 4$     ◀ one part

**STEP 4**    Kath =  $8 \times 4 = 32$     ◀ her share

**ANSWER** 32 goals✓ Arjun 8, Simon 20:  $20 - 8 = 12$  ✓**WATCH OUT**✗  $12 \div 15$  per part✓  $12 \div (5 - 2) = 4$ 

12 is a difference, so divide by the difference in parts, not the total.

**EXAM TIP**

"Write down all the stages": show the 3 parts, the 4 per part, then  $8 \times 4$ .

**Check yourself**

calculator first • answers on page 15

**3.1** Practice • Easy

153 cm of string cut in the ratio 4 : 2 : 3. Each length?

Answer: .....

**3.2** Practice • Easy

Share £630 between Daniel and Rose in the ratio 5 : 13.

Answer: .....

**3.3** Dr Eslam

Boys : girls = 5 : 3.  
There are 18 more boys than girls. How many students?

Answer: .....

**V4****VARIANT 4** MOST COMMON IN THE EXAM

## Linking Two Ratios

Make the shared quantity the same number in both ratios, then join them.

SEEN IN PAST PAPERS

● ● ● ● ● **2 / 5**

PRACTICE QUESTIONS: 2


  
Your notes

Came up in: May 2021 P1H Q4 • Nov 2023 P2H Q5

### ? What is it?

#### In plain words

Two ratios share a

**middle quantity:** flour : butter  
= 6 : 5 and butter : sugar = 3 : 2.

Butter is 5 in one and 3 in the other. Scale both so butter is the **LCM**, 15. Then join: **18 : 15 : 10**.

#### IT LOOKS LIKE

$$a : b = 3 : 4$$

$$b : c = 6 : 5 \Rightarrow a : b : c$$

$$6 : 5 \quad (\times 3) \rightarrow 18 : 15$$

$$3 : 2 \quad (\times 5) \rightarrow 15 : 10$$

$$18 : 15 : 10$$

MATCH THE MIDDLE: LCM OF 5 AND 3

### How do I do it?

#### 1 Middle

Find the quantity in both ratios.

#### 2 LCM

Make it the same number (LCM) in both.

#### 3 Join

Write one three-part ratio.

#### 4 One part

Use the given amount  $\div$  its parts; multiply.

**MATCH THE MIDDLE** 2 of the 5 past papers.

**V4** Linking Two Ratios

VARIANT 4 · SEE IT DONE, THEN TRY


  
Your notes
 WHEN YOU SEE → YOU DO

"chickens : ducks = 7 : 2, ducks : pigs = 5 : 9" → **Ducks to 10:** 35 : 10 : 18

"She uses 120 grams of sugar" → 120 ÷ **sugar's parts** = **one part**

"Write  $a : b : c$  in its simplest form" → **Match the middle, then simplify**

 WORKED EXAMPLE **PAST PAPER**

November 2023 · Paper 2H · Q5 · 3 marks

flour : butter = 6 : 5 and butter : sugar = 3 : 2. Avril uses 120 grams of sugar. Work out the weight of flour she uses.

**STEP 1** 6 : 5 ( $\times 3$ ) = 18 : 15, 3 : 2 ( $\times 5$ ) = 15 : 10 ◀ butter to 15

**STEP 3** flour : butter : sugar = 18 : 15 : 10 ◀ join

**STEP 4** 1 part =  $120 \div 10 = 12$ ,  $18 \times 12 = 216$  ◀ one part, then flour

**ANSWER** 216 g

✓ 216 : 180 : 120 = 6 : 5 and 3 : 2 ✓

 **WATCH OUT**

✗ 6 : 5 : 2

✓ 18 : 15 : 10

Butter can't be 5 and 3 at once — match it first.

 **EXAM TIP**

Check: divide your joined ratio back to both given ratios.

 **Check yourself**

calculator first · answers on page 15

**4.1**

May 2021 P1H

chickens : ducks = 7 : 2,  
ducks : pigs = 5 : 9.  
There are 36 pigs. How  
many chickens?

Answer: .....

**4.2**

Dr Eslam

$a : b = 3 : 4$  and  
 $b : c = 6 : 5$ . Write  
 $a : b : c$  in its simplest  
form.

Answer: .....

**4.3**

Dr Eslam

$x : y = 2 : 3$  and  
 $y : z = 4 : 5$ . Find  
 $x : y : z$ .

Answer: .....

**V5****VARIANT 5** MOST COMMON IN THE EXAM**Ratio Problems**

Split the story into steps; each step is a ratio, a fraction or a one-part calculation.

SEEN IN PAST PAPERS

●●●●● **2 / 5**

PRACTICE QUESTIONS: 19



Your notes

Came up in: Jan 2020 P1H Q5 • Jun 2023 P2H Q5

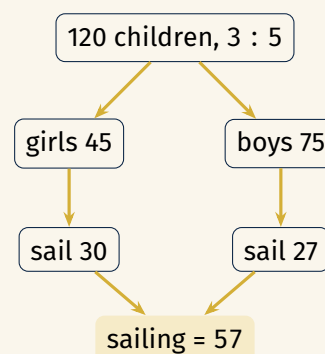
**? What is it?****In plain words**

A longer story mixes ratio with **fractions**, **money values** or a **change**. Do it one step at a time and label each number.

With coins, a **group** is one set of the ratio: 1 : 3 means one 2p and three 5p = 17p per group.

**IT LOOKS LIKE** $\frac{16}{25}$  of the boys

85p in 2p : 5p = 1 : 3

**How do I do it?****1 Split**

Use the first ratio to get the groups.

**2 Step on**

Apply each fraction or ratio to the right group.

**3 Label**

Write what each number means (girls sail, boys sail ...).

**4 Answer**

Combine what the question asks for.

**ONE STEP, ONE LABEL** 5–6 marks: method marks for each step.

**V5 Ratio Problems**

VARIANT 5 • SEE IT DONE, THEN TRY

  
 Your notes
**WHEN YOU SEE → YOU DO***" $\frac{16}{25}$  of the boys go climbing"*→ **Fraction of that group only***Coins with a value and a ratio*→ **Value of one group; total ÷ group****WORKED EXAMPLE** **PAST PAPER**

January 2020 • Paper 1H • Q5 • 6 marks

120 children; girls : boys = 3 : 5.  $\frac{16}{25}$  of the boys go climbing. Twice as many girls go sailing as go climbing. Everyone either sails or climbs. How many children go sailing?

**STEP 1**  $120 \div 8 = 15$  : girls 45, boys 75      ◀ split 3 : 5

**STEP 2** boys climb  $\frac{16}{25} \times 75 = 48$ , sail 27      ◀ fraction of the boys

**STEP 3+4** girls sail  $\frac{2}{3} \times 45 = 30$ ;  $27 + 30 = 57$       ◀ sail : climb = 2 : 1; add
**ANSWER** 57 children✓ climb  $48 + 15 = 63$ ;  $57 + 63 = 120$  ✓**WATCH OUT**

$\times \frac{16}{25} \times 120$

$\checkmark \frac{16}{25} \times 75$

The fraction is of the boys, not of everyone.

**EXAM TIP**

Check the end: sailing + climbing must give back 120.

**Check yourself**

calculator first • answers on page 15

**5.1**

Jun 2023 P2H

85p in 2p and 5p coins;  
 $2p : 5p = 1 : 3$ . How  
 many more 5p coins?

Answer: .....

**5.2**

Dr Eslam

Red : blue = 2 : 3. Add  
 8 red: now 1 : 1. How  
 many blue?

Answer: .....

**5.3**

Dr Eslam

Ages 2 : 3 now, 3 : 4 in  
 5 years. Ages now?

Answer: .....

## CHECK YOURSELF

# Answers



Your notes

Every answer below was checked with sympy (check\_answers.py).

**1.1** 270, 60, 165, 45 g**1.2** 500 g needed: No**1.3** 20 biscuits**2.1**  $n = 20$ **2.2** 1 : 2**2.3** 2 : 3 : 5**3.1** 68, 34, 51 cm**3.2** Daniel £175, Rose  
£455**3.3** 72**4.1** 70 chickens**4.2** 9 : 12 : 10**4.3** 8 : 12 : 15**5.1** 10 more**5.2** 24 blue**5.3** 10 and 15

## THE WHOLE TOPIC ON ONE PAGE

# Ratio cheat sheet


  
Your notes
**V1 Recipes & scaling**

$$\times \frac{\text{new}}{\text{old}}$$

Same multiplier for every amount.

**V2 Simplify & 1:n**

$$2 : 500 = 1 : 250$$

Same units; divide all parts.

**V3 Sharing 1/5**

$$12 \div (5 - 2) = 4$$

Total  $\leftrightarrow$  sum; difference  $\leftrightarrow$  difference.**V4 Linking 2/5**

$$18 : 15 : 10$$

Match the middle (LCM), join.

**V5 Problems 2/5**

$$45, 75 \rightarrow 30 + 27 = 57$$

One step, one label.

**TOP 3 LOST MARKS**

1. dividing a difference by the total  $\times$
2. joining ratios without matching  $\times$
3. mixed units  $\times$

**BOTTOM LINE** Ratio = parts • one part first • match the middle • one step, one label.